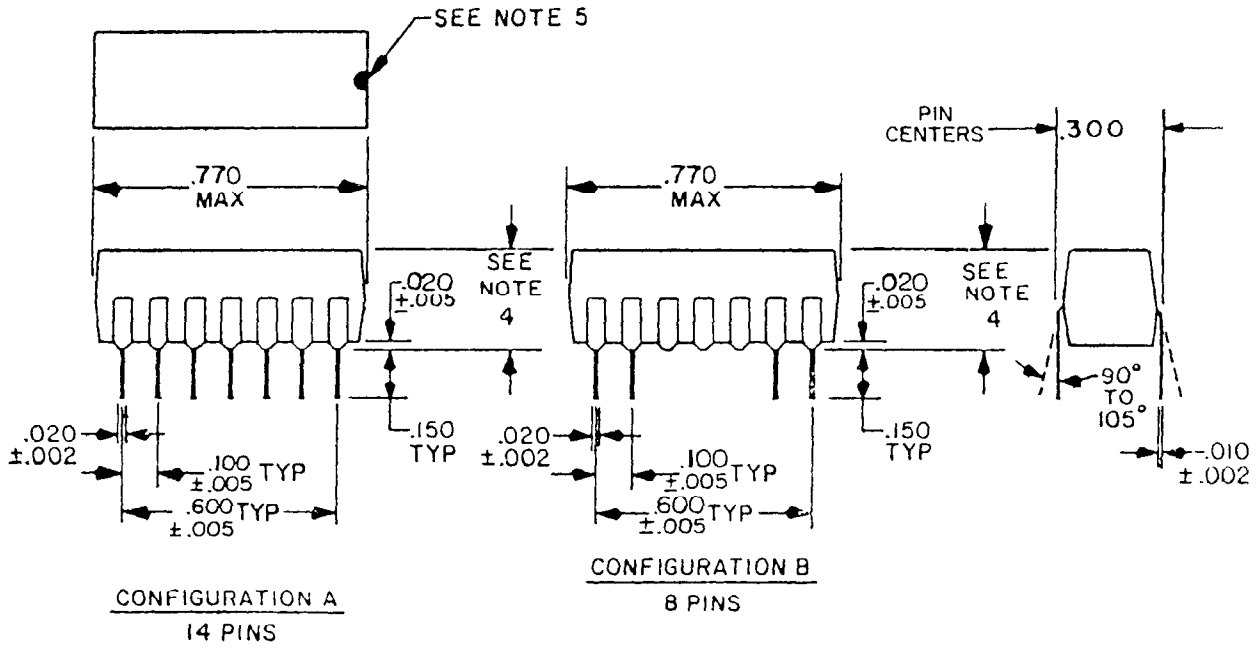
INCH-POUND
MIL-R-83516/1C(USAF)
22 December 1989
SUPERSEDING
MIL-R-83516/1B(USAF)
24 May 1985

MILITARY SPECIFICATION SHEET

RELAYS, REED, DRY, DUAL IN-LINE PACKAGE (DIP), GENERAL PURPOSE

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the relays described herein shall consist of this specification and the latest issue of MIL-R-83516.



INCHES	MM	INCHES	MM
.002	0.05	.100	2.54
.005	0.13	.150	3.81
.010	0.25	.300	7.62
.015	0.38	.600	15.24
.020	0.51	.750	19.05

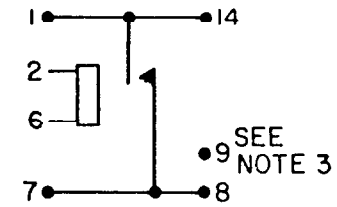
© FIGURE 1. Outline drawing.

© denotes changes

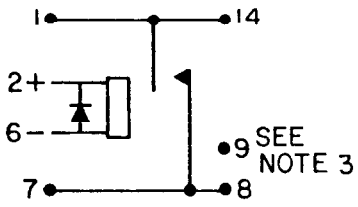
ANSC N/A
FSC 5945

DISTRIBUTION STATEMENT A. Approved for public release: distribution is unlimited.

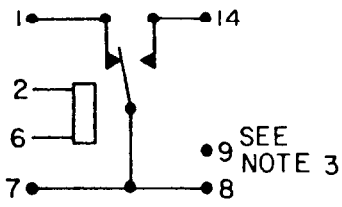
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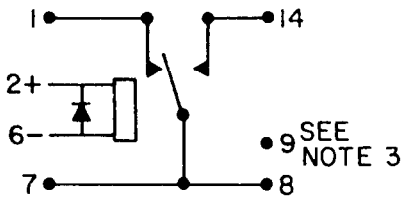
CIRCUIT DIAGRAM A
1 FORM A



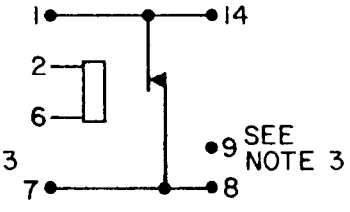
CIRCUIT DIAGRAM B
1 FORM A



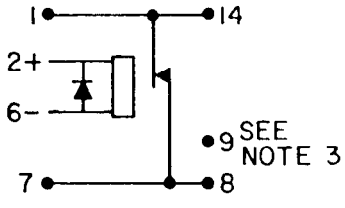
CIRCUIT DIAGRAM C
1 FORM C



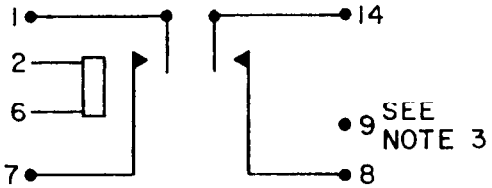
CIRCUIT DIAGRAM D
1 FORM C



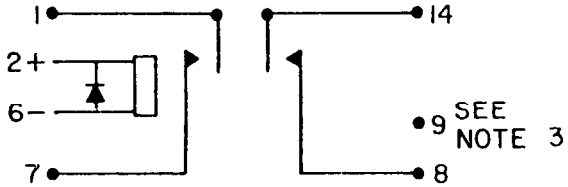
CIRCUIT DIAGRAM E
1 FORM B



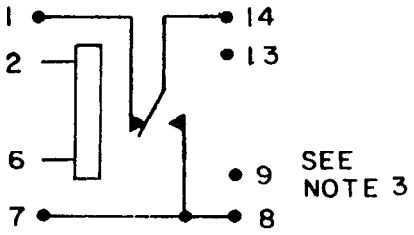
CIRCUIT DIAGRAM F
1 FORM B



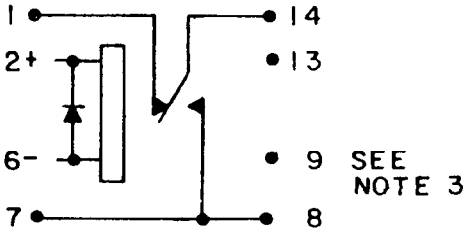
CIRCUIT DIAGRAM G
2 FORM A



CIRCUIT DIAGRAM H
2 FORM A



CIRCUIT DIAGRAM J
1 FORM C



CIRCUIT DIAGRAM K
1 FORM C

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Relays are available with electrostatic shielding at pin 9.
4. Relay heights can be found in table I.
5. A position identifier (shape optional) shall be placed on the top of each relay either centered between pins 1 and 14 or directly over pin 1.
© 6. Unless otherwise specified, tolerance is ±.010(0.25mm)

FIGURE 1. Outline drawing - Continued.

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REQUIREMENTS:

DESIGN AND CONSTRUCTION:

Dimensions and configurations: See figure 1.

Diode characteristics (when applicable): Diodes shall be JANTX as a minimum. 1/

Peak inverse voltage: 75.0 V dc.

CONTACT DATA:

Arrangements: See figure 1.

Load ratings:

Contacts	V dc (max)	mA (max)	VA (max)
Form A	100	500	10
Form B	100	500	10
Form C	28	250	3

Contact resistance:

Before life: 0.130 ohm maximum.

During and after life: 1.0 ohm maximum.

For group A inspection: 0.200 ohm maximum.

Contact noise: Applicable.

Contact stability: Applicable.

COIL DATA: See table I.

Duty rating: Continuous.

Coil power dissipation: See table I.

Operate time (including bounce): See table II.

Release time (including bounce): See table II.

TABLE II. Operate and release times (including bounce).

Contact arrangement	Operate time (nominal)		Release time (nominal)	
	Without diode	With diode	Without diode	With diode
	ms	ms	ms	ms
Form A	1	1	.5	1
Form B	.5	.5	.5	.5
Form C	2	2	2.5	2.5

1/ WARNING: Reverse polarity on coil terminals will destroy diode.

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TABLE I. Dash numbers and applicable characteristics.

Dash number 1/						Circuit diagram (see fig. 1)	Configuration (see fig 1.)	DC coil data at 25°C												Relay height (maximum) inch
Without shielding			With shielding					Resistance (minimum) ohms			Pickup voltage (maximum)			Dropout voltage (minimum)			Power dissipation (nominal) mW			
Coil voltage			Coil voltage					Coil voltage			Coil voltage			Coil voltage			Coil voltage			
5	12	24	5	12	24			5	12	24	5	12	24	5	12	24	5	12	24	
001	002	003	004	005	006	A	A	324	1,296	5,184	3.5	8.25	16.5	0.75	1.8	3.6	70	100	100	.290
007	008	009	010	011	012	B	A	"	"	"	"	"	"	"	"	"	"	"	"	"
013	014	015	016	017	018	A	B	"	"	"	"	"	"	"	"	"	"	"	"	"
019	020	021	022	023	024	B	B	"	"	"	"	"	"	"	"	"	"	"	"	"
025	026	027	028	029	030	C	A	90	450	1,800	"	"	"	"	"	"	250	285	285	"
031	032	033	034	035	036	D	A	"	"	"	"	"	"	"	"	"	"	"	"	"
037	038	039	040	041	042	C	B	"	"	"	"	"	"	"	"	"	"	"	"	"
043	044	045	046	047	048	D	B	"	"	"	"	"	"	"	"	"	"	"	"	"
049	050	051	052	053	054	E	A	112.5	"	"	"	"	"	"	"	"	200	"	"	.375
055	056	057	058	059	060	F	A	112.5	"	"	"	"	"	"	"	"	200	"	"	.375

See footnote at end of table.

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TABLE I. Dash numbers and applicable characteristics - Continued.

Dash number 1/						Circuit diagram (see fig. 1)	Configuration (see fig 1.)	DC coil data at 25°C												Relay height (maximum) inch
Without shielding			With shielding					Resistance (minimum) ohms			Pickup voltage (maximum)			Dropout voltage (minimum)			Power dissipation (nominal) mW			
Coil voltage			Coil voltage					Coil voltage			Coil voltage			Coil voltage			Coil voltage			
5	12	24	5	12	24			5	12	24	5	12	24	5	12	24	5	12	24	
061	062	063	064	065	066	E	B	112.5	450	1,800	3.5	8.25	16.5	0.75	1.8	3.6	200	285	285	.375
067	068	069	070	071	072	F	B	"	"	"	"	"	"	"	"	"	"	"	"	"
073	074	075	076	077	078	G	A	"	"	"	"	"	"	"	"	"	"	"	"	"
079	080	081	082	083	084	H	A	"	"	"	"	"	"	"	"	"	"	"	"	"
085	086	087	088	089	090	G	B	"	"	"	"	"	"	"	"	"	"	"	"	"
091	092	093	094	095	096	H	B	"	"	"	"	"	"	"	"	"	"	"	"	"
097	098	099	100	101	102	J	A	"	"	"	"	"	"	"	"	-	"	"	"	.290
103	104	105	106	107	108	K	A	"	"	"	"	"	"	"	"	"	"	"	"	"
109	110	111	112	113	114	J	B	"	"	"	"	"	"	"	"	-	"	"	"	"
115	116	117	118	119	120	K	B	"	"	"	"	"	"	"	"	-	"	"	"	"

1/ For dash numbers 097-120, the normally closed contacts shall not reclose at 120% rated coil voltage.

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ELECTRICAL DATA:

Dielectric withstanding voltage: MIL-STD-202, method 301. 2/

Test points:

Between all mutually insulated components: 500 V dc.

Across contacts: 250 V dc.

Insulation resistance: 10,000 MΩ minimum.

Test potential: 100 V dc ±10%.

Test points: Between contacts.

Coil transient suppression (relays with internal diodes): 1.0 V maximum negative transient.

Capacitance, pF maximum:	1A	2A	1B	1C
Across open contacts, no shield:	1.25	1.25	1.25	2.0
Across open contacts, shield guarded:	0.25	0.25	0.25	1.2
Between contact and coil, no shield:	3.0	3.0	3.0	3.5
Between contact and coil, shield guarded:	2.0	2.0	2.0	2.0

ENVIRONMENTAL DATA:

Thermal EMF: Not applicable.

Thermal shock: Applicable.

Shock (specified pulse): Applicable.

Vibration, high frequency: Applicable.

Salt spray (corrosion): Applicable.

Moisture resistance: Applicable.

Magnetic interference: Relays are subject to interaction with stray magnetic fields. To keep within operating parameters, relays should be mounted no closer than .500 inch (12.7 mm) of each other. Greater spacing is required for stronger magnetic fields.

Resistance to soldering heat: Applicable.

PHYSICAL:

Enclosure: Molded plastic.

Operating temperature range: -40°C to +105°C.

Lead integrity: MIL-STD-883, method 2004, test condition B2; the force to be applied shall be 8.0 ±0.5 ounces.

Solderability: Applicable.

Life: Applicable.

2/ For relays supplied with an internal diode, the coil leads shall be connected together to avoid damage to the diode.

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Weight (maximum grams):

1 FORM A:	2.5
1 FORM C:	2.5
1 FORM B:	3.3
2 FORM A:	3.0

Part number: M83516/1-(dash number from table I).

QUALIFICATION:

Qualification inspection and sample size: See table III.

TABLE III. Qualification inspection and sample size.

M83516/1-	Number of samples
023	13 units each part number plus 1 open unit.
047	Qualification inspection as applicable.
071	
095	
119	
022 and 024	2 units each part number. Qualification
046 and 048	inspection group I and life.
070 and 072	
094 and 096	
118 and 120	

Custodian:
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

Agent:
DLA - ES

Preparing activity:
Air Force - 85

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